

so that

$$v^\top \nabla^2 L_i(\mathbf{W}_{i*})v = \sum_j \hat{p}_{ij} v_j^2 - \left(\sum_j \hat{p}_{ij} v_j \right)^2 \geq \left(\sum_j \hat{p}_{ij} v_j^2 \right) \min_j \hat{p}_{ij} \geq \min_j \hat{p}_{ij}^2$$

which is $\Theta(\varepsilon^2)$ due to (18). It now follows from classical convex analysis that

$$\begin{aligned} \|\hat{p}_{i*}^{(T_1+T)} - p_{i*}^\varepsilon\| &\leq \|\mathbf{W}_{i*}^{(T_1+T)} - \mathbf{W}_{i*}^*\| \\ &\leq (1 - \Omega(\varepsilon^2))^T \|\mathbf{W}_{i*}^+ - \mathbf{W}_{i*}^*\| \\ &= \exp(-\Omega(\varepsilon^2 T)) \cdot O(\log \varepsilon^{-1}), \end{aligned}$$

where we have again used that softmax is 1-Lipschitz and the effective learning rate $\mu_i \eta = \Theta(1)$. $\square$

D.2 Analysis of Search and PPO

We first show that the initial cluster exploration phase of the search algorithm is consistent.

Lemma D.2. *For each outer loop of Algorithm 2, after T_0 steps, $\hat{C}$ returns the cluster C_k containing X_0 with probability $1 - \tilde{O}(K^{-2})$.*

Proof. $\hat{C} \neq C_k$ implies that either some state $y \in C_k$ has not been hit by some $X_t^{n,\varepsilon}$ by time T_0 , or all chains $X_t^{n,\varepsilon}$ have reached some point outside C_k at time T_0 .

Denote the hitting time of $C \subseteq S$ by $X_t^{n,\varepsilon}$ as $\tau_C^{n,\varepsilon}$. Since $K \leq \text{poly}(M)$, by Lemma B.9 it holds that

$$\mathbb{P}_{X_0}(\tau_y^{n,\varepsilon} \geq T_0) \leq \frac{1}{MK^2}, \quad \forall y \in C_k$$

by choosing $T_0 \geq \Omega(M(\log M)^2)$. Union bounding over y, n gives

$$\mathbb{P}_{X_0} \left(\max_{y \in C_k} \max_{n \leq N} \tau_y^{n,\varepsilon} > T_0 \right) \leq MN \cdot \mathbb{P}_{X_0}(\tau_y^{1,\varepsilon} > T_0) \leq O\left(\frac{\log K}{K^2}\right).$$

Moreover by the argument in Lemma C.12, since each $X^{n,\varepsilon}$ is independently generated from p^ε ,

$$\begin{aligned} \mathbb{P}_{X_0} \left(\max_{n \leq N} \tau_{C_k^c}^{n,\varepsilon} \leq T_0 \right) &\leq \mathbb{P}_{X_0} \left(\tau_{C_k^c}^{1,\varepsilon} \leq T_0 \right)^N \\ &\leq \left(1 - \left(1 - \Theta\left(\frac{\varepsilon}{M}\right) \right)^{T_0} \right)^N \\ &\leq \exp \left(-N \exp \left(-\Theta\left(\frac{T_0 \varepsilon}{M}\right) \right) \right) \\ &\leq \exp(-N/2) \leq K^{-2} \end{aligned}$$

by ensuring that $N \geq 4 \log K$. Hence we have shown that $\mathbb{P}_{X_0}(\hat{C} \neq C_k) \leq \tilde{O}(K^{-2})$. $\square$

Proposition D.3 (Proposition 3.3 restated). *PRM mode of Algorithm 2 returns $\mathcal{M}_s = E_s$ with probability $1 - \tilde{O}(1/K)$.*

Proof. Denote the set of outbound edges from C_k as $E_{s,k} := \{(x, y) \in E_s : x \in C_k, y \in C_k^c\}$ so that $|E_{s,k}| \leq n_{\text{out}} d_{\text{out}}$ and fix $(x, y) \in E_{s,k}$. The probability that a fixed rollout $X^{1,\varepsilon}$ takes the edge (x, y) and terminates within time $T_{\max}$ is bounded below as

$$\begin{aligned}
& \mathbb{P}_{X_0} (\exists t \leq T_{\max} : X_{t-1}^{1,\varepsilon} = x, X_t^{1,\varepsilon} = y) \\
&= \sum_{t=1}^{T_{\max}} \mathbb{P}_{X_0} (\tau_{C_k^c}^{1,\varepsilon} \geq t, X_{t-1}^{1,\varepsilon} = x, X_t^{1,\varepsilon} = y) \\
&\geq \sum_{t=1}^{T_{\max}} \mathbb{P}_{X_0} (\tilde{X}_{t-1}^{k,\varepsilon} = x) \mathbb{P}_{X_0} (\tau_{C_k^c}^{1,\varepsilon} \geq t) p^\varepsilon(y|x) \\
&\geq \sum_{t=2t_{\text{mix}}}^{T_{\max}} \frac{\rho}{2M} \left(1 - \Theta\left(\frac{\varepsilon}{M}\right)\right)^{t-t_{\text{mix}}} \Theta(\varepsilon) \\
&\geq \left(1 - \Theta\left(\frac{\varepsilon}{M}\right)\right)^{t_{\text{mix}}} \left(1 - \left(1 - \Theta\left(\frac{\varepsilon}{M}\right)\right)^{T_{\max}-2t_{\text{mix}}+1}\right) \\
&\geq \exp\left(-O\left(\frac{\varepsilon \log M}{M}\right)\right) \cdot \exp\left(-\exp\left(-\Theta\left(\frac{T_{\max}\varepsilon}{M}\right)\right)\right) \geq c,
\end{aligned}$$

for some positive constant c . Therefore by union bounding over all edges in $E_{s,k}$,

$$\begin{aligned}
\mathbb{P}_{X_0} (\hat{E} \neq E_{s,k}) &\leq \sum_{(x,y) \in E_{s,k}} \mathbb{P}_{X_0} (\nexists t \leq T_{\max} : X_{t-1}^{1,\varepsilon} = x, X_t^{1,\varepsilon} = y)^N \\
&\leq |E_{s,k}| (1 - c)^N \leq O(1/K^2)
\end{aligned}$$

by taking $N/\log K$ suitably large. By union bounding again over all $R = \Theta(K \log K)$ iterations of the outer loop, the probability that the inner loop fails to return the correct set of sparse edges for some iteration is at most $\tilde{O}(1/K)$. Finally, the probability that some cluster will be missed during the R iterations is bounded above for a suitable choice of R as

$$K (1 - \Theta(1/K))^R \leq K \exp(-\Theta(R/K)) \leq O(1/K).$$

Thus with probability $1 - \tilde{O}(1/K) - O(1/K)$, all clusters C_k will be explored and the correct set of sparse edges $E_{s,k}$ will always be added to $\mathcal{M}_s$, so that the final output satisfies $\mathcal{M}_s = E_s$. $\square$

We now prove the convergence of the PPO-Clip algorithm.

Proof. As in the proof of Proposition 3.3, we condition on the event that the correct set of sparse edges is returned for all clusters, and focus on a single cluster C_k . Again write

$\hat{p}_{ij}^{(t)} = \hat{p}_{\mathbf{W}^{(t)}}(e_j|e_i)$, let $X_0 \sim \mu$ where $\mu_i = \Theta(KM)$ and denote $D_{s,i} = \{j \in [S] : (i, j) \in E_s\} \subset D_i$. The objective L_{PPO} reduces to $\sum_{i \in C_k} \mu_i L_i$ where

$$L_i(\mathbf{W}; \hat{A}) = \sum_j p_{ij}^\varepsilon \min \left\{ \frac{\hat{p}_{\mathbf{W}}(e_j|e_i)}{p_{ij}^\varepsilon}, c_{\text{clip}} \right\} \hat{A}(e_i, e_j) = \sum_{j \in D_{s,i}} \min \{ \hat{p}_{\mathbf{W}}(e_j|e_i), c_{\text{clip}} \cdot p_{ij}^\varepsilon \}.$$

We will show inductively that either $\hat{p}_{\mathbf{W}}(e_j|e_i) < c_{\text{clip}} \cdot p_{ij}^\varepsilon$ for all $j \in D_{s,i}$ or $\hat{p}_{\mathbf{W}}(e_j|e_i) \geq c_{\text{clip}} \cdot p_{ij}^\varepsilon$ for all $j \in D_{s,i}$ while running Algorithm 3. Assuming the former, we see that for all $j \in [S]$,

$$\begin{aligned} \frac{dL_i}{dw_{ij}}(\mathbf{W}^{(t)}; \hat{A}) &= \frac{d}{dw_{ij}} \sum_{k \in D_{s,i}} \hat{p}_{ik}^{(t)} = \frac{d}{dw_{ij}} \sum_{k \in D_{s,i}} \frac{\exp w_{ik}^{(t)}}{\sum_\ell \exp w_{i\ell}^{(t)}} \\ &= 1_{\{j \in D_{s,i}\}} \frac{\exp w_{ij}^{(t)}}{\sum_\ell \exp w_{i\ell}^{(t)}} - \sum_{k \in D_{s,i}} \frac{\exp w_{ik}^{(t)} \cdot \exp w_{ij}^{(t)}}{(\sum_\ell \exp w_{i\ell}^{(t)})^2} \\ &= \hat{p}_{ij}^{(t)} \left(1_{\{j \in D_{s,i}\}} - \sum_{k \in D_{s,i}} \hat{p}_{ik}^{(t)} \right). \end{aligned}$$

This implies that $\frac{dL_i}{dw_{ij}}(\mathbf{W}^{(t)}; \hat{A}) > 0$ if and only if $j \in D_{s,i}$, so sign gradient descent gives for each i the update rule

$$w_{ij}^{(t+1)} = \begin{cases} w_{ij}^{(t)} + \mu_i \alpha & j \in D_{s,i} \\ w_{ij}^{(t)} - \mu_i \alpha & j \notin D_{s,i}. \end{cases}$$

In particular, the relative magnitudes of $\hat{p}_{ij}^{(t)}$ for all $j \in D_{s,i}$ are preserved starting from $\hat{p}_{ij}^{(0)} = p_{ij}^\varepsilon$, so that the earlier assertion is justified. Furthermore since $\hat{p}_{ij}^{(t)} \propto \exp w_{ij}^{(t)}$ for all j , we can derive $\hat{p}_{ij}^{(t)}$ from $\hat{p}_{ij}^{(0)}$ by directly multiplying $e^{\pm \mu_i \alpha}$ (or $e^{2\mu_i \alpha}$ and 1) and normalizing afterwards,

$$\tilde{p}_{ij}^{(t)} := \begin{cases} e^{2\mu_i \alpha t} p_{ij}^\varepsilon & j \in D_{s,i} \\ p_{ij}^\varepsilon & j \notin D_{s,i} \end{cases} \quad \text{and} \quad \hat{p}_{ij}^{(t)} = \frac{\tilde{p}_{ij}^{(t)}}{\sum_k \tilde{p}_{ik}^{(t)}}.$$

By choosing

$$T_i = \frac{1}{2\mu_i \alpha} \log(1 + o(1)) c_{\text{clip}} = \frac{1}{2\mu_i \alpha} \log \frac{\varepsilon_{\max}}{\varepsilon},$$

we ensure that gradient descent has not yet terminated (reached the clip threshold) by time T_i and also $\tilde{p}_{ij}^{(t)} \leq \varepsilon_{\max}$ for all $j \in D_{s,i}$. This implies

$$1 \leq \sum_j \tilde{p}_{ij}^{(t)} \leq 1 + (e^{2\mu_i \alpha t} - 1) \sum_{j \in D_{s,i}} p_{ij}^\varepsilon \leq 1 + (e^{2\mu_i \alpha t} - 1) |D_{s,i}| \varepsilon \leq 1 + O(\varepsilon_{\max}).$$